PRESDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

MONDULI DISTRICT COUNCIL

MANYARA SECONDARY SCHOOL

FORM ONE ANNUAL EXAMIMINATION

031 **PHYSICS**

Name …… Stream …….

TIME: 2.30 HOURS **NOVEMBER, 2025**

Instructions

1) This paper consists of three sections A, B and C with total of ten (10) questions.
2) All answer must be written in space provided.
3) Where necessary use the following constants
   i. Acceleration due to gravity (g) = $10m/s^2$
   ii. Density of water $1g/cm^3$ or $1000Kg/m^3$

SECTION A (15 Marks)

Answer **all** questions in this section

1. For each of the items (i)-(x), choose the correct answer from the given alternatives and write its letter beside the item number in the space provided. **(10marks)**

(i) Which of the following is a set of equipment used in school that use the knowledge of physics.

A Sawmills, cranes and spanners
B Fax machines, air planes and train
C Computers, printers and projectors
D X-ray, ultrasound and syringe

(ii) Which set best represents fundamental physical quantities with their SI units?

A Length (km), mass (kg), time (s), temperature (℃), amount of substance (mol) and electric current (A)
B Length (m), mass (kg), time (s), electric current (I), temperature (℃) and amount of substance (mol)
C Length (m), mass (kg), time (s), temperature (K), electric current (A) and luminous intensity (Cd)
D Length (km), mass (g), time (s), temperature (K), electric current (A) and amount of substance (mol)

(iii) Which one represent the fundamental forces?

A Compressional, torsional and frictional forces
B Gravitational, torsional and frictional forces
C Nuclear, gravitational and electromagnetic forces

D Stretching, attraction and repulsion forces

(iv) The mass of 25 cm^3 of ivory was found to be 0.045 kg. The density of ivory in kg/m^3 is

A 1.8 kg/m^3 B 180 kg/m^3 C 18000 kg/m^3 D 1800 kg/m^3

(v) Why does a piece of steel sink in water but a steel ship floats?

A. The density of the steel ship is less than the density of water

B. Steel is denser than the steel ship

C. Steel ship has the same density to that of steel

D. The average density of steel ship is less than the density of water

(vi) During a study tour at Mwalimu Nyerere Hydroelectric power station, Mr. Mbise wondered to see the walls of a dam are made thicker at the bottom than at the top. As a physicist, the reason behind for this is……

A. Pressure of water at the bottom is less
B. Pressure in liquid decrease with the increase of depth
C. Pressure of water at the bottom is greater
D. Weight of water at the bottom is greater

(vii) What is the relationship between pressure and area?

A On decreasing area, pressure decreases
B On increasing area, pressure increases
C On decreasing area, pressure increases
D On changing area, pressure remains constant

(viii) A body is uniformly accelerated from rest to a final velocity of 100 m/s in 10 seconds. The distance covered is

A 10 m B 1000 m C 110 m D 500 m

(ix) What happens when the body moves with a constant speed?

A Its acceleration is zero
B Its acceleration increases
C Its acceleration decreases
D Its deceleration increases

(x) What is the kinetic energy of a 10 kg object moving with a 2 m/s?

A 10 J B 20 J C 40 J D 100 J

Qn	I	ii	Iii	Iv	V	Vi	Vii	Viii	ix	X
Answer										

2. Match the descriptions in LIST A with the name of its corresponding concept related to linear motion in LIST B by writing a letter of the correct response beside the corresponding item number in the answer sheet provided. **(05 marks)**

LIST A	LIST B
(i) Used to measure gas pressure (ii) Uses Pascal's principle to operate (iii) Used to measure atmospheric pressure (iv) Application of atmospheric pressure (v) Application of pressure in solid	A. Minimum pressure B. Depend on depth of the container C. Maximum pressure D. Aneroid barometer E. Razor blades, knife F. siphon G. Hydraulic press H. Manometer I. Pascal J. At high mountain

List A	i	Ii	Iii	iv	V
List B					

SECTION B (70 Marks)

Answer **all** questions in this section

3. (a) What is meant by the following terms

(i) Dependent variable (**1 marks)**

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(ii) Independent variable **(1 marks)**

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(iii) Controlled variable **(1 marks)**

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(b) A student is investigating the relationship between number of coils in a spring and the distance it compresses when a force is applied. He vary the number of coils and measure the compression distance. Determine the dependent, independent and controlled variables in this experiment **(3 marks)**

(i) Dependent variable

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(ii) Independent variable

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(iii) Controlled variable

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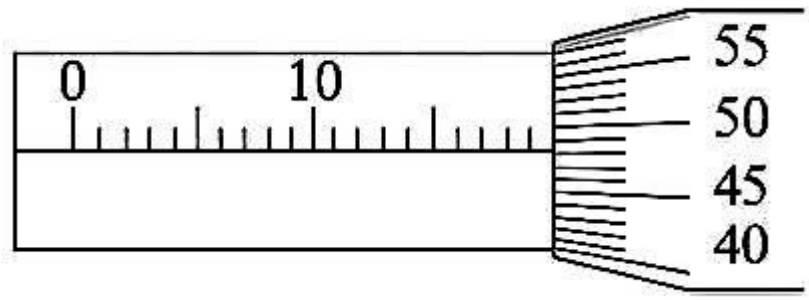
(c) The rocket moves from the earth to planet Z. If it weights 5000N and 200N on the earth and on planet Z respectively, determine the gravitational force on planet Z. **(4 Marks)**

4. (a) Distinguish the following terms with three points. **(3 marks)**

Mass	Weight

(b) What is the reading of the micrometer screw in the figure below

(3 marks)



(c) State four effects of force caused by unbalanced force acting on an object. **(4 marks)**

(i)……...
(ii)……
(iii)……
(iv)……

5. (a) You as a form one Physics expert at Manyara secondary school, identify the two conditions for the work to be done. **(2 Marks)**

(i)……

(ii)……

(b) (i) Differentiate between the two types of mechanical energy. **(3 Marks)**

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(ii) Explain Why a rotating wonder wheel becomes hot after a sudden stop. **(2 Mark)**

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(c) A model of a car of mass 20 kg moves at a speed of 6 m/s for 10 seconds. Find the power developed by its engine. **(3 marks)**

6. (a) Give two differences between density and relative density **(3 marks)**

Density	Relative density

(b) If a 200 g solid lead is placed in a eureka can filled with water, what volume of water will overflow? Take the density of lead as 11.3 g/cm^3 **(3 Marks)**

(c) In an experiment to determine the relative density of a liquid, a solid Q weighted as follows: **(4 Marks)**

Weight in air, $w_0 = 8.6$ N

Weight in water, $w_1 = 6.0$ N

Weight in Liquid, $w_2 = 5.4$ N.

Calculate the relative density of the liquid.

7. (a) (i) A science class is learning about the properties of materials and observes that some objects sink in water, while others float, even though they are made of the same material. Suggest and state the principle related to the above scenario. **(2 marks)**

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(ii) Explain why a fat person can learn to swim easily than a slim one? **(2 Marks)**

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(b) A stone weighs 2 N in air and 1.2 N when totally immersed in water.

Calculate:

(i) The upthrust **(2 marks)**

(ii) The volume of the stone **(2 marks)**

(iii) The density of the stone **(2 marks)**

8. (a) (i) Differentiate between Capillarity and Surface tension. **(2 Marks)**

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(ii) Explain why hot soup is tastier than a cold soup? **(1.5 Marks)**

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(iii) When water and mercury were in two separate measuring cylinders, the teacher asked student to observe the reading in both cylinders. Why does mercury form downward meniscus while that of water forms upward meniscus? **(2 Marks)**

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(b) (i) During an activity in class, each student stretches a helical spring and observes its behavior of elasticity upon releasing it. Suggest and state the principle of physics to which this phenomenon obey. **(2 Marks).**

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(ii)A piece of wire of length 12 cm is stretched through 2.5 cm by a mass of 50 N. Assuming that the wire is elastic material, what force will stretch it through 4 cm?

(2.5 marks)

9. (a) (i) Define the term pressure and suggest its SI unit **(2 marks)**

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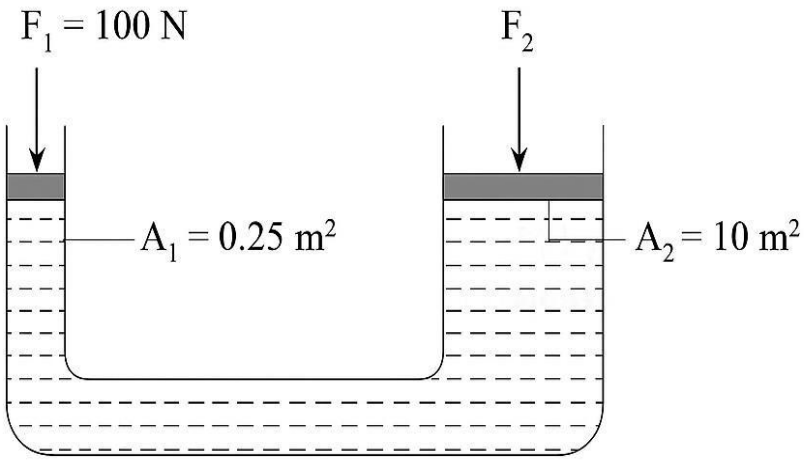
(ii) A group of students are observing plastic bottle containing water. They realize that when bottle is punctured and squeezed, water squirts from all holes with the same pressure. Suggest and state the principle related to this observation. **(2 marks)**

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(iii) The atmospheric pressure is the result of the total weight of the air layers on the Earth's surface. Briefly explain why does not crush us as we know its value is about 1.03×10^5 Pa? **(2 Marks)**

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(b) A small force of 100 N applied on the small piston of areas A_1 equal to 0.25 m^2 produces a bigger force F_2 on a larger piston of area A_2 equal to 10 m^2. Calculate F_2 **(4 marks)**



SECTION C (15 Marks)

Answer question **ten (10)**

10. (a) (i) What is the difference between acceleration and retardation **(2 marks)**

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(ii) Write three equations of uniformly accelerated motion **(3 marks)**

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(b) (i) If the coin and a feather are released from the same height at the same time in the vacuum, which one will reach the ground first? Explain **(2 marks)**

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(ii) A falling object is pulled down by the earth. The earth is pulled up towards the object. Why doesn't the earth move? **(2 Marks)**

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(c) A car starts from rest and accelerates uniformly at the rate of 4 m/s^2 for 5 seconds. It maintained a constant velocity for 20 seconds. The brakes are applied and the car decelerates for 3 seconds.

(i) Determine the maximum velocity attained. **(2 Marks)**

(ii) Draw the velocity time graph. **(2 Marks)**

(iii) Find the total distance covered. **(2 Marks)**